

Transportable Relational Patient Profiles for Interpretable Clustering of Omics Data

Acronym: TRPP Scheme: SONATA BIS 16 Duration: 48 months Primary panel: ST6 Document status: complete scientific draft; administrative, personnel, and budget data must be supplied by the applicant before submission.

1. Scientific objective

Patient stratification from omics measurements depends on an early decision that is usually implicit: what does it mean for two patients to be similar? Distances between normalised expression values, distances between within-sample ranks, and distances between pairwise ordering relations such as gene A > gene B define different geometries and can yield different groups. There is no reason to expect one geometry to be best for every disease, platform, or cohort shift. Moreover, a stable partition of a discovery cohort does not by itself imply biological relevance or transportability to new patients.

TRPP will develop a leakage-controlled framework that first audits whether a source cohort supports value, relational, hybrid, or no stable structure and then learns short relational descriptions of supported groups. A complete source artifact - preprocessing, selected features and relations, distance scales, representation decision, profiles, thresholds, and assignment rule - will be frozen before it is applied to independent cohorts. Target samples will be assigned without reclustering, joint normalisation, or target tuning; an uncertain sample may remain UNASSIGNED.

The central hypothesis is that biologically useful patient groups can be represented by sparse sets of stable within-patient relations, provided that representation adequacy is assessed on source data only and the applicability domain is explicit. Four objectives make this hypothesis falsifiable:

1. Representation adequacy. Establish when VALUE, RELATIONAL, HYBRID, or NO_STABLE_STRUCTURE is supported by source-only diagnostics.
2. Direct relational regions. Develop a deterministic algorithm that learns assignments and concise core/optional relational profiles jointly, and compare it with relational PAM followed by post-hoc rule extraction.
3. Frozen transport. Test whether source-learned profiles retain coverage, stability, interpretability, and external association in at least two target cohorts per disease module without target refitting.
4. Applicability map. Identify signal, noise, missing-feature, confounding, and platform-shift conditions under which relational information helps, is neutral, or fails.

The project does not assume that relational representations always win. A valid outcome is a map showing that value geometry is preferable in some regimes and that grouping should be withheld in others.

2. Significance and state of the art

High-dimensional omics studies commonly apply clustering after feature filtering and normalisation. Yet algorithm choice is only part of the problem: representation determines which variation is treated as signal. PAM provides a useful controlled engine because it accepts arbitrary dissimilarities and uses observed samples as medoids, but it still returns a partition for data with little meaningful cluster structure. Stability resampling and prediction strength address whether partitions persist under perturbation, while adjusted Rand and mutual information quantify agreement after labels are revealed. These tools do not, however, provide a general source-only decision among candidate data geometries or a frozen rule system for assigning future samples.

Within-sample relative expression orderings have a complementary history in supervised molecular classifiers. Top-scoring-pair methods showed that a few comparisons can generate transparent decisions and reduce dependence on absolute scale. Their success does not establish an unsupervised theory: clustering lacks outcome labels during learning, pair spaces grow quadratically, and technical shifts can alter gene-specific orderings. Conversely, conventional batch correction can be effective for pooled analyses but is incompatible with the primary frozen-transfer question when target distributional information is used to refit the source representation.

TRPP connects four elements that are normally studied separately:

- an audit of representation adequacy before profile learning;
- explicit abstention when stability does not exceed a matched null process;
- direct learning of sparse regions made of within-sample relations rather than retrospective explanation of an arbitrary partition;
- application of a frozen patient-profile artifact to independent cohorts, including rejection when evidence is insufficient.

The project's novelty is therefore not the use of ranks, pair comparisons, PAM, or stability alone. It is the auditable architecture and the direct optimisation of transportable relational regions under a strict source/target boundary. Each important claim has a negative counterpart: no stable structure, no advantage over a post-hoc baseline, insufficient target feature coverage, or no retained external association. These outcomes remain reportable rather than being repaired after labels are inspected.

Preliminary work supports the question but does not settle it. A deterministic Python pilot compared value, rank-relational, and hybrid representations using one PAM implementation. In 630 controlled source-target pairs, the source-only audit identified the generating family in 93.3% of signal replicates; median target ARI regret was 0.000, the NULL false-structure rate was 6.7%, HYBRID was never selected in pure regimes, and the source-audit/target-performance Spearman association was 0.854. Gate B: GO.

The real-data evidence was deliberately more cautious. In eleven frozen within-dataset audits, the selected representation was within 0.05 ARI of the retrospective oracle in 9/11 cases, but median agreement with available labels was only 0.065. In bidirectional GSE10072/GSE19804 transfer, one direction matched the oracle (ARI 0.926; regret 0.000), whereas the reverse direction obtained ARI 0.559 versus oracle 0.664 (regret 0.105284), exceeding the frozen 0.10 allowance. Gate C: STOP. Thresholds were not relaxed; direct regions: NOT TESTED; anchors: NOT TESTED. The labels were used only after all source-only decisions and assignments had been frozen. This mixed result identifies the precise gap addressed by TRPP: within-cohort adequacy must be connected to an explicit, multi-cohort model of transportability.

3. Concept and work plan

The 48-month project is organised into four overlapping work packages (WPs) with preregistered transitions.

WP1. Representation adequacy and abstention (months 1-12)

WP1 will extend the controlled benchmark to include additive and multiplicative noise, gene-specific offsets, monotone transformations, missing features, platform mapping errors, nuisance structure, unequal groups, and unknown cluster number. VALUE, RELATIONAL, and HYBRID will use the same deterministic PAM in the matched primary comparison. Source-only audit quality will combine perturbation stability, prediction strength, non-degeneracy, and performance above matched NULL distributions. Its output is a ranked adequacy record or NO_STABLE_STRUCTURE, not a promise that the chosen partition matches an unseen phenotype.

WP1 gate: profile learning proceeds only for non-degenerate solutions whose source quality exceeds the frozen NULL eligibility rule. The controlled benchmark must retain acceptable family discrimination, false-structure control, and low target regret under held-out regimes. Failure narrows the audit to descriptive diagnostics rather than triggering target-driven recalibration.

WP2. Direct sparse relational regions (months 7-24)

WP2 will first implement RR_POSTHOC: deterministic relational PAM followed by profile extraction. It will then develop RR_DIRECT, which alternates between deterministic assignment and selection of relations that define compact regions. A relation can be positive, tied, or negative; missingness is preserved rather than silently converted to a tie. Profiles will distinguish core relations, expected in most members, from optional relations that increase coverage without dominating assignment. Candidate lengths of 10, 25, and 50 relations will be declared in advance. Redundant inverse relations and excessive reuse of a single gene will be controlled. All tie-breaking, empty-group handling, and stopping conditions will be recorded in a trace.

WP2 gate: relative to RR_POSTHOC, RR_DIRECT must improve external ARI/NMI by at least 0.05, improve bootstrap profile Jaccard by at least 0.10, or remain within 0.03 while producing a materially shorter profile. Otherwise the direct method stops and the post-hoc profile remains the supported result.

WP3. Frozen multi-cohort transfer (months 18-38)

The lung module will use GSE10072 as the default source and GSE19804 and GSE32863 as targets. It combines demographic/etiological and platform shifts; GSE18842 is a preregistered broader-NSCLC sensitivity cohort. The colorectal module will use tumour profiles from GSE39582 as source and GSE14333 and GSE33113 as targets. CMS, stage, molecular markers, and outcomes are evaluation-only; CMS is acknowledged as expression-derived rather than an independent truth.

Each task will produce a hashed prelabel artifact. No target cohort will select features, relations, cluster number, weights, or rejection thresholds. Insufficient mapping produces an ineligible task or UNASSIGNED, not a new target-optimised profile. Primary baselines are value PAM, relational PAM, HYBRID PAM, and RR_POSTHOC; conventional cluster-then-classify procedures are secondary comparators.

WP4. Generalisation map and conditional efficiency (months 31-48)

WP4 will integrate controlled and real-cohort results into a domain-of- applicability map. It will explicitly model negative cases: stable nuisance, label-discordant structure, relation flips, low coverage, group collapse, and poor assignment margins. Only after the WP2 gate passes will one restricted single-anchor experiment test computational efficiency. Anchor sets are not in the core scope. Release artifacts will include Python code, manifests, frozen configurations, synthetic generators, and auditable result tables.

4. Methodology and evaluation

Data boundary and reproducibility

Every dataset is represented by a `DatasetBundle` with measurements and sample metadata separated from evaluation labels. Fit and audit packages cannot import the label loader. Source preprocessing includes quality control, feature mapping, transformation, robust scaling where required, and median absolute deviation feature selection. The fitted artifact is serialised with ordered feature identifiers, parameters, software versions, seeds, and hashes. Target processing is a pure application of that artifact. A regression test will alter all target values or labels and confirm byte-identical source artifacts.

All primary algorithms are deterministic, CPU-only, and Python-first. Average ranks handle ties; ternary relations encode -1, 0, and +1; missing values remain explicit. Value Euclidean/correlation, rank footrule, pair-Hamming, and hybrid distances are tested for symmetry, zero diagonal, boundedness where applicable, tie behaviour, and deterministic byte output. A golden small-matrix test verifies the global PAM optimum, and permutation tests verify canonical tie-breaking.

Representation Audit

For each candidate representation and source-only split, the audit estimates partition stability, prediction strength, minimum cluster fraction, non-degeneracy, and margin above a matched permuted/noise null. Hybrid weights are a finite preregistered grid and are normalised from source distances only. The audit returns a quality score with components retained separately so that a high aggregate cannot conceal a degenerate group. Selection and cluster number are frozen before evaluation labels are available.

Direct regions and assignment

`RR_DIRECT` will optimise a transparent objective balancing within-region relation agreement, between-region contrast, profile length, redundancy, and coverage. Optimisation is deterministic and bounded by a declared candidate relation budget. A new sample is compared with every frozen profile; assignment requires sufficient observable relations and a source-calibrated margin over the next profile. Otherwise it is `UNASSIGNED`. No target sample can change a profile or the assignment of another target sample; individual-versus-batch equivalence is a required test.

Outcomes and analysis

Controlled outcomes are representation-family accuracy, target ARI/NMI, regret versus a retrospective oracle, NULL false-structure rate, unjustified HYBRID selection, and association between source audit differences and target behaviour. Real-cohort outcomes are assignment coverage, `UNASSIGNED` rate, minimum group size, relation coverage and flip rate, profile-state retention, assignment margin, profile length, gene count, bootstrap Jaccard, and external ARI/NMI where a suitable reference exists. Clinical and pathway associations are evaluated only after freezing and are reported with effect sizes, uncertainty intervals, and multiplicity control.

Uncertainty will be estimated by resampling independent patients; paired specimens and technical replicates will remain grouped. Comparisons will report all preregistered directions and datasets, not only favourable results. A hierarchical summary across cohorts will supplement, not replace, per-cohort results. Sensitivity analyses will cover cluster-number uncertainty, feature-mapping rules, missing relations, and histological mismatch. No post-label adjustment can change a primary result.

Expected outputs

The expected scientific outputs are: a tested representation-adequacy and abstention framework; a direct relational-region algorithm or a clearly documented negative result; multi-cohort frozen-transfer evidence in two disease contexts; and an empirical map defining when relational profiles are credible. The software output will be an open, deterministic reference implementation, not a clinical device or web portal.

5. Team, feasibility, risk, and impact

The new team separates method construction from independent validation. The PI will define the mathematical and experimental architecture, supervise all WPs, and integrate the resulting applicability theory. A doctoral researcher, employed for at least the scheme-required 36 months, will develop RR_DIRECT, its correctness tests, and stability analysis as a coherent dissertation. A postdoctoral researcher or specialist analyst will lead cohort curation, feature mapping, reproducibility review, and frozen validation. External domain consultation will be named only after formal confirmation. Individual names and achievements will be inserted from verified CV and institutional records before submission.

Feasibility is supported by a completed deterministic pilot: a common PAM engine, source-only preprocessing artifacts, label import firewall, four distance families, controlled generators, real-data adapters, frozen transfer, and 138 automated tests at pilot closeout. The full controlled benchmark and all decisions have machine-readable evidence. Crucially, the failed external gate was preserved rather than repaired, showing that the proposed stopping rules are operational.

The main risks and responses are:

- Stable nuisance resembles biology. Technical perturbations, metadata diagnostics, matched nulls, and independent cohorts are used; stability alone never establishes meaning.
- The audit ranks candidates but cannot choose reliably. The output becomes an adequacy profile with abstention rather than a universal selector.
- Direct regions do not beat post-hoc profiles. The WP2 gate stops the new method and retains RR_POSTHOC as the supported result.
- Cross-platform relation coverage is low. The source-defined mapping and threshold are retained; low-coverage samples remain unassigned, and the failure defines an applicability boundary.
- Profiles are stable but label-discordant. Structural and phenotype validity are reported separately; labels cannot rescue training.
- Quadratic relation space is expensive. Source-only screening, redundancy control, and a fixed candidate budget bound CPU cost. A single-anchor restriction is tested only conditionally; anchor sets remain out of scope.
- A result depends on one cohort or direction. At least three cohorts per disease module and complete directional reporting reduce selection bias.

TRPP will contribute a disciplined answer to a practical but under-specified question in omics clustering: when is a patient grouping defensible and portable? Its broader impact is methodological. It treats representation, abstention, interpretability, and transfer as linked objects and supplies an audit trail that allows reviewers and future users to distinguish learned evidence from evaluation labels. The project is basic research; it will not claim clinical utility without separate prospective validation.

6. References

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